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Appendix 1. PubMed search strategy

1. neovascular Glaucoma as Topic [Mesh]
2. ((glaucoma* or angle* or iris or anterior) near/4 (neovascular*)) [tw]
3.((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*)) [tw]
4. (NVG or NVI or NVA) [tw]
5. 1 or 2 or 3 or 4
6. randomized controlled trial [pt] or controlled clinical trial [pt] or randomized [tiab] or placebo [tiab] or clinical trials as topic [Mesh] or randomly [tiab] or trial [ti] or retrospective
7. animals [mh] not humans [mh]
8. 5 and 6 not 7

Appendix 2. Embase search strategy

1. neovascular glaucoma/exp
OR
2. ((glaucoma* or angle* or iris or anterior) adj4 neovascular*):ab
3. ((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) adj4 glaucoma*):ab
4. (NVG or NVI or NVA):ab
5. 1 or 2 or 3 or 4
6. [controlled clinical trial]/lim or [randomized controlled trial]/lim or retrospective
7. 5 and 6

Appendix 3. Central (Cochrane) search strategy

1. MeSH descriptor: [Glaucoma, Neovascular] explode all trees
2. (glaucoma* or angle* or iris or anterior) near/4 (neovascular*)
3. (haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*)
4. NVG or NVI or NVA
5. 1 or 2 or 3 or 4

Appendix 4. Web of Science search strategy

1. “Neovascular Glaucoma”
2. ((glaucoma* or angle* or iris or anterior) near/4 (neovascular*)) [tw]
3. ((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*))[tw]
4. NVG or NVI or NVA
5. 1 or 2 or 3 or 4

Appendix 5 References list of included studies

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3 . M. W. Zhou, W. Wang, W. B. Huang, et al. Adjunctive with versus without intravitreal bevacizumab injection before ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. Chinese Medical Journal 2013; 126(8): 1412-1417.
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8 . F. Kong, X. Ma, S. Fan, et al. Efficacy of intravitreal lucentis or conbercept injection combined with Ahmed glaucoma valve implantation for treatment of neovascular glaucoma. Journal of Jilin University Medicine Edition 2017; 43(6): 1237-1242.
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12 . X. L. Zhang, S. Y. Li, Z. P. Zhang, et al. Clinical effect of trabeculectomy and glaucoma valve implantation on neovascular glaucoma. International Eye Science 2014; 14(2): 349-351.
13 . J. T. Sun, H. J. Liang, M. An, et al. Efficacy and safety of intravitreal ranibizumab with panretinal photocoagulation followed by trabeculectomy compared with Ahmed glaucoma valve implantation in neovascular glaucoma. Int J Ophthalmol 2017; 10(3): 400-405.
14 . L. Gao,W. Liu. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. Zhonghua Shiyan Yanke Zazhi/Chinese Journal of Experimental Ophthalmology 2018; 36(8): 613-618.
15 . Q. W. Du,L. S. Yang. Effects of intravitreal injection of Bevacizumab combined with complex trabeculectomy for advanced neovascular glaucoma. International Eye Science 2015; 15(10): 1766-1768.
16 . F. Wang,L. L. Wang. Safety of anti-VEGF drugs with trabeculectomy for neovascular glaucoma. International eye science 2016; 16(5): 837-840.
17 . Y. Saito, T. Higashide, H. Takeda, et al. Beneficial effects of preoperative intravitreal bevacizumab on trabeculectomy outcomes in neovascular glaucoma. Acta Ophthalmologica 2010; 88(1): 96-102.

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Appendix 6 Success and failure criteria applied in each study

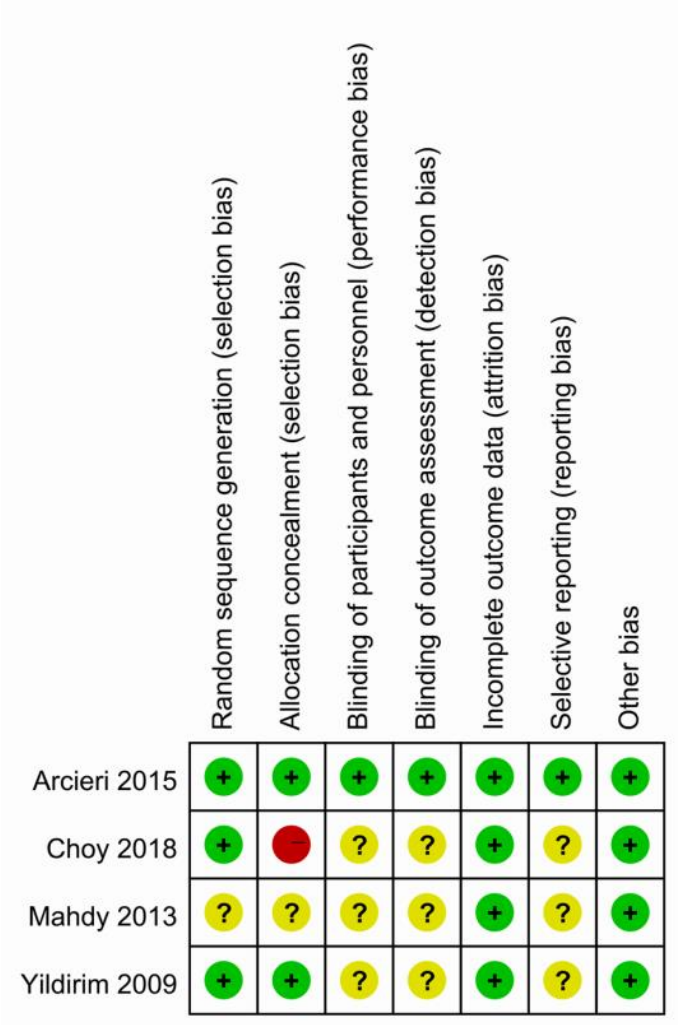
References	Outcomes
Ma,2012	- A failure of operation was defined by: (1) IOP more than 21mmHg using a Goldmann application despite the use of antiglaucoma medication at 2 consecutive visits, (2) light perception-negative vision (except when preoperative vision acuity was light perception negative), (3) when additional glaucoma surgery was needed, or (4) operative or postoperative devastating complications, such as endopthalmitis, were noted.
Mahdy,2013	- Complete success was defined as an IOP≤21mmHg and ≥10mmHg without antiglaucoma medications or additional glaucoma surgery and without visually devastating complications such as endophthalmitis or phthisis bulbi and no loss of light perception. - Qualified success was considered when the above criteria were fulfilled, but with additions of antiglaucoma topical medical treatment. - Failure of shunt surgery in the study included instances when the above criteria of complete or qualified success were not fulfilled, a lack of control of IOP with or without glaucoma medications, operative or postoperative devastating complications, loss of light perception, or the need for additional glaucoma surgical intervention.
Zhou,2013	- Postoperative survival was defined as IOP < 21mmHg, with or without glaucoma medications, and without significant vision-threatening complications (endophthalmitis, retinal detachment, suprachoroidal hemorrhage, preseptal cellulitis, or persistent hypotony (IOP < 5mmHg)). - A complete success was defined as IOP between 6 and 21mmHg without any antiglaucoma medication. - Qualified success was defined as IOP between 6 and 21mmHg with one medication or more. - Failure was defined as IOP of more than 21mmHg despite the use of maximum-tolerated medications; IOP < 5mmHg on two consecutive follow-up visits after 3 months; or the need for additional glaucoma surgery or laser to control IOP, removal of the shunt implant, or serious complication.
Kang,2014	- Surgical success was defined as an IOP≤21 mmHg with or without anti-glaucoma medication, and maintenance of visual acuity of light perception or better (except when preoperative visual acuity was NPL). - Surgical failure was defined as an IOP≥22 mmHg at two consecutive follow-up visits, or when additional glaucoma surgery was needed. Postoperative use of anti-glaucoma medication was not a criterion of success or failure.
Arcieri,2015	- Success was defined by the following criteria: (1) post-operative IOP level between 6 and 21 mmHg, with or without antiglaucoma medications, (2) IOP reduction of at least 30% relative to preoperative values. - Eyes requiring additional glaucoma surgery that developed phthisis or that showed loss of light perception were classified as failures.
Kwon,2017	- Surgical success was defined as an IOP between 6 and 21 mmHg, without loss of light perception (LP), and with or without the use of anti-glaucoma medication. - Surgical failure was defined as an IOP of more than 21 mmHg or less than 6 mmHg at two consecutive follow-up visits, the loss of LP, or a need for additional glaucoma interventions.

Li,2019	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of anti-glaucoma medication, and without serious ocular complication. - Qualified success was considered as an IOP between 6 and 21 mmHg, with 1 or 2 kinds of anti-glaucoma medication. - Failure was defined as: (1) IOP more than 21 mmHg or less than 6 mmHg, with 3 kinds of anti-glaucoma medication, (2) when additional glaucoma surgery was needed, (3) light perception-negative vision when preoperative vision acuity was light perception negative, (4) serious ocular complications were mentioned, or (5) removal of the shunt implant. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Kong,2017	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of antiglaucoma medication, and without serious ocular complication. - Qualified success was defined as an IOP between 6 and 21 mmHg, with one medication or more. - Failure was defined as: (1) IOP more than 21 mmHg at two consecutive follow-up visits, with 3 kinds of anti-glaucoma medication or more, and a need for additional glaucoma surgery; (2) IOP less than 6 mmHg at consecutive follow-up visits; or (3) serious complications or (4) removal of the shunt implant due to ocular complications. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Yildirim,2009	- Surgical success was defined as IOP less than 21mmHg and greater than 5mm Hg without additional glaucoma surgery and without loss of light perception. Postoperative use of antiglaucoma medications was not accepted as a criterion of success or failure. The definition of hypotonia in this study was IOP of 5mm Hg or less in more than 2 consecutive visits. - The patients who underwent repeated DCPC and needling procedures were considered as treatment failures.
Choy,2018	- In the TSCP group, the laser procedure was repeated as necessary, spaced at least 4 weeks apart, until either the IOP was 21 mmHg or below, or a total of five treatments had been given. Uncontrolled IOP requiring any additional, non-assigned interventions was counted as failure. Additional procedures in relation to the implanted AGV (for example, clearing tube blockage, repair of implant exposure, excision of encapsulated bleb, and repositioning of implant) were not counted as failure but documented as complications. Similarly, any TSCP-related conjunctival burn or scleral thinning, with or without need for additional procedures, was recorded as a complication. - Successful IOP control was defined as a final visit IOP not higher than 21 mmHg, with or without medication, and no hypotony-related maculopathy or choroidal detachment. - Successful IOP control excluding the 1 TsCP and 5 agV eyes that did not follow the assigned protocol. - Overall success is defined as not only IOP normalization but also preservation or improvement of BCVA. Overall success = Valid cases – (failed IOP control + deteriorated BCVA).
Sun,2017	- Complete success was defined by the following criterion: postoperative IOP between 6 and 21 mm Hg without anti-glaucoma medications and without further anti-glaucoma surgery. - The qualified success criterion was IOP between 6 and 21 mmHg with one or more medications. - Failure was defined as IOP>21 mm Hg despite the use of maximum medications. If a patient maintained stable IOP after an additional surgery because of a complication, this was not considered failure of the procedure; however, the additional surgery could not be an

	anti-glaucoma surgery. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Gao,2018	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of anti-glaucoma medication, and without serious ocular complication. - Qualified success was defined as an IOP between 6 and 21 mmHg, with one medication or more, and without serious ocular complication. - Failure was defined as: (1) IOP more than 21 mmHg at two consecutive follow-up visits, with 3 kinds of anti-glaucoma medication or more, and a need for additional glaucoma surgery; (2) IOP less than 6 mmHg at consecutive follow-up visits, with serious complications or (4) removal of the shunt implant due to ocular complications. - Serious complications included: (1) Drainage tube related complications: exposure, displacement, obstruction and corrosion of drainage tube; (2) Non-drainage tube related complications: retinal detachment, corneal decompensation, malignant glaucoma, supracoroidal hemorrhage, endophthalmitis, long-term low intraocular pressure. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Du,2015	- Surgical success was defined as an IOP between 7 and 22 mmHg, according to the general efficacy criteria.
Wang,2016	- Surgical success was defined as an IOP between 7 and 22 mmHg, according to the general efficacy criteria.
Sisto,2007	- Successes and failures were defined at the last post-operative visit. - Complete success was defined as post-operative IOP < 21 mmHg without medical therapy. - Qualified success was defined as post-operative IOP < 21 mmHg with medical therapy. - Failure was defined as postoperative IOP≥21 mmHg, despite maximal medical treatment, along with cases demonstrating no light perception(NLP) postoperatively.
Saito,2010	- The surgery was considered to have been successful when IOP < 21 mmHg with or without medication (qualified or complete success, respectively) was achieved. - A procedure was considered to have failed if IOP exceeded these criteria at two consecutive follow-up visits (follow-up visits were made monthly up to 1 year and once every several months thereafter) or in the presence of phthisis bulbi, loss of light perception or need for additional glaucoma surgeries, including surgical revision, trabeculectomy and cyclophotocoagulation.
Takihara,2011	- The surgical success and failure were defined before the data analysis. Surgical success was defined as an IOP ≤21 mmHg with or without topical ocular hypotensive medication and the maintenance of visual acuity of light perception or better. - Surgical failure was defined as an IOP continuously ≥22 mmHg at 2 consecutive follow-up visits, a deterioration of visual acuity to NPL, or additional glaucoma surgeries (such as filtering surgery and destruction of the ciliary body). We completed laser suture lysis within 2 months after MMCT. An IOP ≥22 mm Hg within 2 months after MMCT was not considered as a surgical failure because laser suture lysis was not completed.
Lee,2015	- Surgical success was defined as an IOP less than 21 mmHg, with or without anti-glaucoma medication and the maintenance of visual acuity of light perception or better. - Surgical failure was defined as more than 22 mmHg, a deterioration of visual acuity to NPL, or additional glaucoma surgeries (such as filtering surgery).

Yu,2018	- Complete success was defined as an IOP less than 21 mmHg without anti-glaucoma medication. - Qualified success was defined as an IOP between 21 and 26 mmHg without anti-glaucoma medication, or IOP less than 21 mmHg with one medication. - Failure was defined as an IOP more than 30 mmHg, and still more than 21mmHg when using two kinds of medication.
Shen,2011	- Surgical success was defined as 6 mm Hg ≤ intraocular pressure ≤ 21 mm Hg, with or without glaucoma medications, without further glaucoma surgery including cyclophotocoagulation or complications that required removal of the Ahmed implant, and without loss of light perception. - For failure due to elevated intraocular pressure, two consecutive visits with intraocular pressure ≥ 21 mm Hg were required. Laser suture lysis and bleb needling to improve bleb function were not considered failure of the procedure.
Liu,2016	- Complete success was defined as IOP ≥ 6 mmHg and≤ 21 mmHg without any anti-glaucoma medications or further glaucoma surgery, and without loss of light perception. - Partial success was defined as IOP < 21 mmHg with topical anti-glaucoma medicines. - Surgical failure was defined as IOP ≥ 21 mmHg even with anti-glaucoma medicines, or additional urgical treatment was needed to control IOP , or loss of light perception.

Appendix 7 Risk of bias summary of RCT



Note: The yellow circle with question mark represents “unclear risk of bias”, the red one with minus sign represents “high risk of bias” and the green one with plus sign represents “low risk of bias” (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Notes: NOQAS = New Castle – Ottawa Quality Assessment Scale.

Appendix 8 The quality assessment according to the NOQAS of each cohort study

Study	Selection	Comparability	Outcome	Total score
Ma,2012	3	1	3	7
Zhou,2013	4	2	3	9
Kang,2014	3	2	2	7
Kwon,2017	3	2	3	8
Li,2019	3	2	2	7
Kong,2017	3	2	3	8
Liu,2014	3	2	1	6
Zhang,2014	3	2	1	6
Sun,2017	3	0	3	6
Gao,2018	3	2	3	8
Du,2015	3	0	3	6
Wang,2016	3	2	3	8
Saito,2010	3	2	3	8
Takihara,2011	3	2	3	8
Lee,2015	3	2	3	8
Yu,2018	4	2	2	8
Song,2019	4	2	1	7
Shen,2011	3	1	3	7
Liu,2016	3	1	2	6

Notes: NOQAS = New Castle – Ottawa Quality Assessment Scale.

Appendix 9 Pair-wise meta-analysis

Comparison

With statistically significant effect

Mean difference (95%CI) in IOP 6 months after surgery

AGV	4.74 (1.04,8.45)	1.03 (-4.25,6.32)	2.41 (-5.00,9.82)	4.13 (-2.46,10.72)	6.19 (0.99,11.40)
-2.55 (-6.70,1.59)	AGV+IVAV	-3.71 (-10.18,2.76)	-2.33 (-9.25,4.59)	-0.61 (-6.65,5.42)	1.45 (-3.03,5.93)
-3.88 (-11.62,3.85)	-1.33 (-10.11,7.45)	CPC	1.38 (-7.72,10.47)	3.10 (-5.35,11.54)	5.16 (-2.25,12.57)
3.99 (-5.06,13.03)	6.54 (-1.75,14.83)	7.87 (-4.01,19.75)	CCT	1.72 (-4.92,8.36)	3.78 (-1.49,9.05)
2.70 (-5.36,10.75)	5.25 (-2.82,13.32)	6.58 (-4.58,17.74)	-1.29 (-11.44,8.86)	Trab(MMC)	2.07 (-1.98,6.11)
0.79 (-5.45,7.03)	3.34 (-1.74,8.43)	4.68 (-5.24,14.59)	-3.19 (-9.74,3.35)	-1.90 (-9.66,5.85)	Trab(MMC) +IVAV

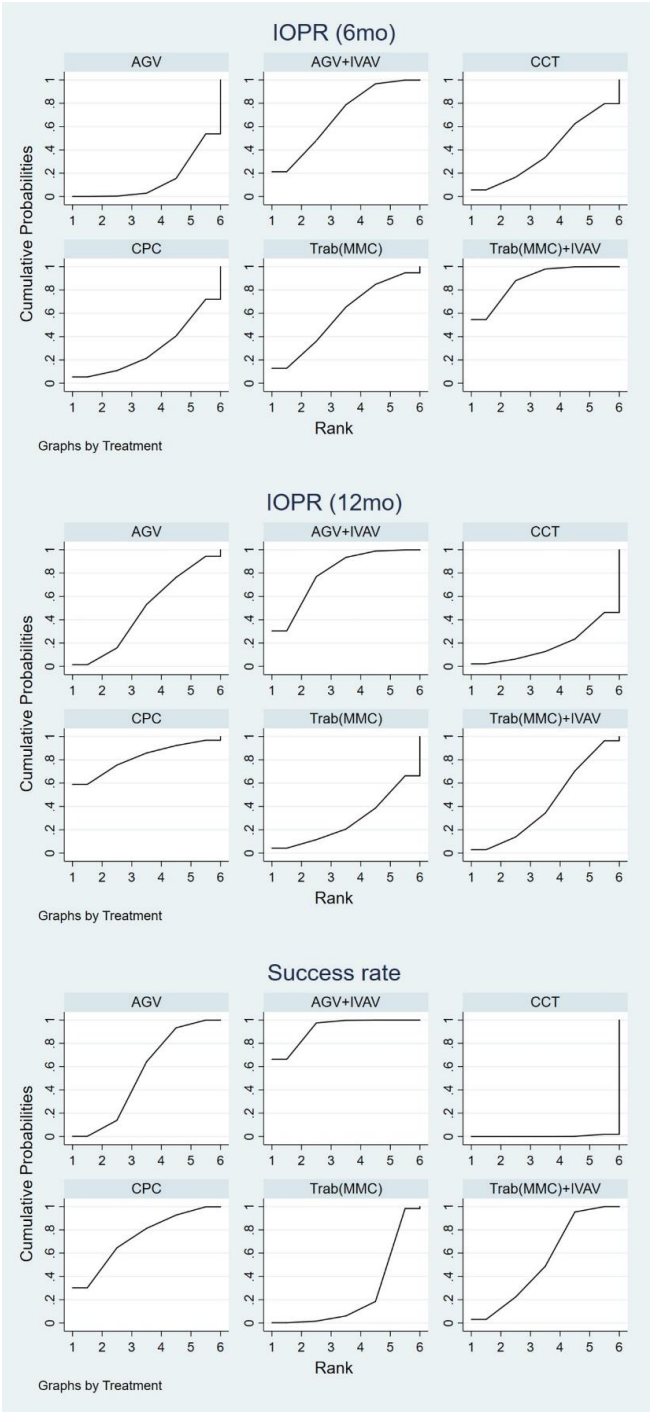
Mean difference (95%CI) in IOP 12 months after surgery

Odds ratio (95%CI) in failure rate

AGV	-0.53 (-0.95,-0.10)	-0.29 (-1.17,0.58)	1.84 (0.65,3.02)	0.57 (-0.27,1.40)	0.08 (-0.65,0.82)
-0.53 (-0.96,-0.11)	AGV+IVAV	0.23 (-0.74,1.21)	2.36 (1.21,3.52)	1.09 (0.27,1.92)	0.61 (-0.08,1.30)
-0.29 (-1.17,0.58)	0.24 (-0.73,1.21)	CPC	2.13 (0.66,3.60)	0.86 (-0.35,2.06)	0.38 (-0.77,1.52)
1.80 (0.63,2.97)	2.33 (1.19,3.48)	2.09 (0.63,3.55)	CCT	-1.27 (-2.40,-0.15)	-1.75 (-2.68,-0.83)
0.54 (-0.29,1.36)	1.07 (0.25,1.89)	0.83 (-0.37,2.03)	-1.26 (-2.39,-0.14)	Trab(MMC)	-0.48 (-1.12,0.16)
0.04 (-0.67,0.76)	0.58 (-0.10,1.25)	0.34 (-0.79,1.47)	-1.80 (-2.97,-0.63)	-0.49 (-1.13,0.15)	Trab(MMC) +IVAV

Odds ratio (95%CI) in success rate

Appendix 10 The surface under the cumulative ranking plots based on cumulative probabilities of interventions



Appendix 11 Node-splitting analysis

Side/Outcomes	Direct		Indirect		Difference		P > |z|	tau
	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error		
IOPR (6mo)								
A B	3.944	2.028	10.178	5.248	-6.234	5.624	0.268	3.241
A F	10.400	4.606	4.115	3.243	6.2847	5.633	0.265	3.241
B F	0.177	2.537	6.424	5.025	-6.247	5.628	0.267	3.241
D F	3.784	2.689	12.388	309.772	-8.604	309.783	0.978	3.251
E F	2.064	2.064	12.451	199.487	-10.387	199.497	0.958	3.251
IOPR (12mo)								
A B	2.550	2.316	2.605	8.314	-0.056	8.631	0.995	4.549
A E	-2.700	5.830	-2.754	6.405	0.054	8.661	0.995	4.551
B F	-3.405	2.917	-3.346	8.142	-0.059	8.649	0.995	4.550
D F	3.195	3.340	-1.584	330.177	4.779	330.193	0.988	4.241
E F	1.900	5.212	1.846	6.918	0.054	8.662	0.995	4.551
Success rate								
A B	0.564	0.227	0.256	0.707	0.308	0.743	0.679	0.000
A E	-0.205	0.641	-0.782	0.553	0.577	0.846	0.495	0.000
A F	-1.185	0.914	0.171	0.398	-1.356	0.997	0.174	0.000
B F	-0.483	0.415	-0.791	0.616	0.308	0.743	0.679	0.000
D F	1.754	0.473	-0.354	95.137	2.109	95.139	0.982	0.000
E F	0.595	0.359	0.018	0.766	0.577	0.846	0.495	0.000
Failure rate								
A B	-0.564	0.227	-0.117	0.744	-0.446	0.778	0.566	0.000
A E	0.205	0.641	0.848	0.566	-0.643	0.855	0.452	0.000
A F	2.197	1.143	-0.171	0.398	2.368	1.210	0.050	0.000
B F	0.483	0.415	0.930	0.658	-0.446	0.778	0.566	0.000
D F	-1.754	0.473	0.435	95.135	-2.189	95.137	0.982	0.000
E F	-0.595	0.359	0.049	0.776	-0.643	0.855	0.452	0.000

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab, trabeculectomy; MMC, mitomycin; IVAV, intravitreal anti-VEGF; IOPR (6mo), intraocular pressure reduction 6 months after surgery; IOPR (12mo), intraocular pressure reduction 12 months after surgery; A=AGV, B=AGV+IVAV, C=CPC, D=CCT, E= Trab(MMC), F= Trab(MMC)+IVAV.

Appendix 12 GRADE for outcome measurements

Comparison	N (studies)	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
IOPR (6mo)					Mixed evidence			
01vs 02	4	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
01vs 03	2	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
01vs 06	1	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
02 vs 06	2	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
04 vs 06	2	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
05 vs 06	3	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
IOPR (6mo)					Indirect evidence			
01 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
01 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 03	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
03 vs 06	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
04 vs 05	-	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
IOPR (12mo)					Mixed evidence			
01vs 02	5	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
01vs 03	2	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
01 vs 05	1	Some concerns	Undetected	Some concerns	Some concerns	Some concerns	No concerns	Very low
02 vs 06	3	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
04 vs 06	2	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
05 vs 06	1	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
IOPR (12mo)					Indirect evidence			
01 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
01 vs 06	-	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
02 vs 03	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
02 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 06	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
04 vs 05	-	Some concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low

Notes: N(studies)=number of studies, IOPR (6mo)= intraocular pressure reduction 6 months after surgery, IOPR (12mo)= intraocular pressure reduction 12 months after surgery; 01=AGV, 02=AGV+IVAV, 03=CPC, 04=CCT, 05=Trab(MMC), and 06=Trab(MMC)+IVAV.

(Continue on the previous table)

Comparison	N (studies)	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Success rate				Mixed evidence				
01vs 02	9	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
01vs 03	2	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
01 vs 05	1	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
01vs 06	1	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 06	2	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
04 vs 06	2	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
05 vs 06	4	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
Success rate				Indirect evidence				
01 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 03	-	Some concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
02 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	Some concerns	No concerns	Some concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
03 vs 05	-	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
03 vs 06	-	Some concerns	Undetected	Some concerns	Some concerns	Some concerns	No concerns	Very low
04 vs 05	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate

Appendix 13 Adverse events of included studies

Author Year	Sample size	Corneal edema	Hyphema	Low IOP	Shallow anterior	Vitreous hemorrhage	Choroidal detachment	Retinal detachment	Tube exposure	Tube occlusion	Encapsulated plate	Phthisis bulbi	Bleb leak	Anterior segment inflammation	Eye pain	Ocular atrophy
AGV vs AGV+IVAV																
Ma 2012	32/ 20	1/0	3/1	—	—	4/1	—	1/0	4/2	1/1	—	—	—	—	—	—
Mahdy 2013	20/ 20	2/0	17/4	3/2	6/5	—	1/0	—	1/0	1/0	6/1	1/0	—	—	—	—
Zhou 2013	28/ 25	—	8/1	—	4/4	—	4/3	—	—	—	7/4	—	—	—	—	—
Arcieri 2013	20/ 20	2/1	6/2	—	1/2	—	4/3	1/0	0/1	—	—	—	—	1/0	—	—
Kang 2014	13/ 14	—	—	2/1	0/1	—	1/0	—	0/2	1/2	—	—	0/2	—	—	—
Li 2019	34/ 46	4/4	8/2	2/4	1/2	—	—	—	—	8/2	—	—	—	3/4	1/2	—
AGV vs AGV+IVAV vs AGV+IVAV																

Kong 2017	21/26/21	—	10/9/11	2/3/3	5/6/7	—	—	—	2/4/3	—	2/1/3	—	—	9/7/9	—	—
AGV vs CPC																
Yildirim 2009	33/33	0/2	7/0	1/3	—	—	1/0	—	3/0	—	3/0	2/0	—	0/5	—	—
Liu 2014	31/26	10/12	2/8	—	—	—	—	—	—	—	—	—	—	5/6	7/18	0/2
Choy 2018	12/8	1/0	1/0	—	—	—	—	—	2/0	—	1/0	3/0	—	—	—	—
AGV+IVAV vs Trab(MMC)+IVAV																
Zhang 2014	13/10	—	—	—	2/1	—	—	—	—	—	—	—	—	—	—	—
Sun 2017	23/22	—	3/4	—	3/2	0/2	—	—	—	—	3/0	—	—	—	—	—
Gao 2018	36/40	—	5/3	11/9	8/4	—	4/2	—	—	—	—	—	—	—	—	—
CCT vs Trab(MMC)+IVAV																
Du	23/	15/0	7/2	—	—	—	—	—	—	—	—	—	—	9/0	16/0	—

2015	23															
Wang 2016	30/30	16/0	6/2	—	—	—	0/1	—	—	—	—	—	0/1	7/0	20/0	—
Trab(MMC) vs Trab(MMC)+IVAV																
Saito 2010	32/20	—	20/5	—	1/0	3/2	4/4	—	—	—	—	—	1/4	—	—	—
Takihara 2011	33/24	—	19/5	2/4	—	7/2	5/2	—	—	—	—	2/0	4/0	—	—	—
Lee 2015	26/32	—	9/5	—	—	—	8/13	—	—	—	—	—	—	—	—	—
Yu 2018	42/48	—	4/1	2/2	6/2	—	—	—	—	—	—	—	—	—	—	—
Song 2019	82/88	3/0	7/1	1/2	3/2	—	—	—	—	—	—	—	0/1	0/1	5/2	1/0
AGV vs Trab(MMC)																
Shen 2011	20/20	—	7/8	0/2	—	—	1/1	—	—	—	—	—	0/1	—	—	—
Liu 2016	19/18	—	1/1	6/0	3/0	2/1	—	—	—	2/0	—	—	—	1/0	—	—

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab, trabeculectomy; MMC, mitomycin; IVAV, intravitreal anti- vascular endothelial growth factor; IOP, intraocular pressure reduction.

Appendix 14 Composition of the primary diseases in the included studies

References	Study design	Interventions	DR	RVO	OIS	CRAO	Others	Combine PRP
Ma,2012	OCT	AGV vs AGV+IVAV	26 vs 15	5 vs 5	1 vs 0	—	—	Yes
Mahdy,2013	RCT	AGV vs AGV+IVAV	15 vs 16	4 vs 4	1 vs 0	—	—	Yes
Zhou,2013	OCT	AGV vs AGV+IVAV	12 vs 7	10 vs 13	1 vs 2	2 vs 1	3 vs 2	Yes
Kang,2014	OCT	AGV vs AGV+IVAV	9 vs 12	2 vs 1	—	—	2 vs 1	Yes
Arcieri,2015	RCT	AGV vs AGV+IVAV	11 vs 10	9 vs 10	—	—	—	Yes
Kwon,2017	OCT	AGV vs AGV+IVAV	14 vs 33	3 vs 5	5 vs 7	—	—	Yes
Li,2019	OCT	AGV vs AGV+IVAV	14 vs 20	20 vs 26	—	—	—	N/A
		AGV vs AGV+IVAV vs						N/A
Kong,2017	OCT	AGV+IVAV	10 vs 12 vs12	10 vs 11 vs 7	1 vs 3 vs 2	—	—	
Yildirim,2009	RCT	AGV vs CPC	NA	NA	NA	NA	NA	NA
Liu,2014	OCT	AGV vs CPC	13 vs 12	11 vs 9	—	—	7 vs 5	NA
Choy,2018	RCT	AGV vs CPC	6 vs 4	4 vs 1	1 vs 3	1 vs 0	—	Yes
Zhang,2014	OCT	AGV+IVAV vs Trab(MMC)+IVAV	5 vs 3	5 vs 5	—	—	3 vs 2	Yes
Sun,2017	OCT	AGV+IVAV vs Trab(MMC)+IVAV	—	—	—	—	—	Yes
Gao,2018	OCT	AGV+IVAV vs Trab(MMC)+IVAV	16 vs 16	16 vs 22	3 vs 1	1 vs 1	—	Yes

Du,2015	OCT	CCT vs Trab(MMC)+IVAV	19	17	—	—	10	Yes
Wang,2016	OCT	CCT vs Trab(MMC)+IVAV	14 vs 14	14 vs 13	—	—	4 vs 3	Yes
Saito,2010	OCT	Trab(MMC) vs Trab(MMC)+IVAV	22 vs 14	6 vs 6	4 vs 0	—	—	Yes
Takihara,2011	OCT	Trab(MMC) vs Trab(MMC)+IVAV	28 vs 18	4 vs 6	1 vs 0	—	—	Yes
Lee,2015	OCT	Trab(MMC) vs Trab(MMC)+IVAV	24 vs 25	2 v 4	0 vs 3	—	—	Yes
Yu,2018	OCT	Trab(MMC) vs Trab(MMC)+IVAV	16 vs 18	26 vs 30	—	—	—	Yes
Song,2019	OCT	Trab(MMC) vs Trab(MMC)+IVAV	43 vs 43	39 vs 45	—	—	—	NA
Shen,2011	OCT	AGV vs Trab(MMC)	14 vs 13	4 vs 3	1 vs 1	2 vs 2	—	NA
Liu,2016	OCT	AGV vs Trab(MMC)+IVAV	8 vs 6	9 vs 10	1 vs 0	—	—	NA

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab, trabeculectomy; MMC, mitomycin; IVAV, intravitreal anti- vascular endothelial growth factor; DR, diabetic retinopathy; RVO, retinal vein occlusion; OIS, ocular ischemic syndrome; CRAO, central retinal artery occlusion; PRP, panretinal photocoagulation.